

TE 458 — Coverage Planning: Tutorial 1 Practice Set (Variant C) — Solutions

4 more numeric variants of Tutorial 1's problems, same methods, new numbers

These are freshly parameterised variants of Tutorial 1's four problems — same methods and formulas (see the companion Formula Sheet), different input numbers — for additional self-test practice. Propagation-model coefficients are given directly in each question, exactly as they would be handed to you on an exam.

Problem 1 — 5G NR Downlink at 1.8 GHz

(a) EIRP

$$\text{EIRP} = P_{\text{TX}} + G_{\text{TX}} - L_{\text{TX}} = 46 + 15 - 3$$

EIRP = 58 dBm

(b) Bandwidth

$$\begin{aligned}\text{Noise floor} &= N_0 + 10\log_{10}(W) + \text{NF} \\ 10\log_{10}(W) &= -93 - (-173.98) - 8 = 72.98 \\ W &= 10^{7.298}\end{aligned}$$

W ≈ 19.85 MHz (≈ 20 MHz)

(c) Receiver Sensitivity

$$S_i = \text{Noise floor} + E_b/N_0 = -93 + 6$$

S_i = -87 dBm

(d) Maximum Allowable Path Loss (MAPL)

$$\begin{aligned}\text{MAPL} &= \text{EIRP} - S_i - M_{\text{shadow}} - M_{\text{interference}} \\ \text{MAPL} &= 58 - (-87) - 6 - 3\end{aligned}$$

MAPL = 136 dB

(e) Cell Radius

$$\begin{aligned}\text{MAPL} &= 126.0 + 35.0 \log_{10}(R) \\ \log_{10}(R) &= (136 - 126.0) / 35.0 = 0.2857\end{aligned}$$

R = 10^{0.2857} ≈ 1.931 km

Problem 2 — 5G NR Uplink at 1.8 GHz (Log-Distance Model)

Body-loss convention: as in the original Tutorial 1 Problem 2, body loss is not folded into EIRP here — EIRP only covers TX power, antenna gain and feeder loss (the hardware between the PA and the antenna). Body loss is a propagation-path loss, so it is added later, at the MAPL step, alongside building penetration loss. Contrast this with Problem 3 below.

(a) EIRP

$$\begin{aligned}P_{\text{TX}} &= 10\log_{10}(125) = 20.97 \text{ dBm} \\ \text{EIRP} &= P_{\text{TX}} + G_{\text{TX}} - L_{\text{TX}} = 20.97 + 0 - 0\end{aligned}$$

EIRP ≈ 20.97 dBm

(b) Free-Space Path Loss at $d_0 = 100 \text{ m}$

$$\begin{aligned}
 PL_0 &= 20\log_{10}(f) + 20\log_{10}(d) - 147.56 \\
 &= 20\log_{10}(1.8 \times 10^9) + 20\log_{10}(100) - 147.56 \\
 &= 185.11 + 40.00 - 147.56
 \end{aligned}$$

$$PL_0 \approx 77.55 \text{ dB}$$

(c) Noise Calculations at $T_0 = 295 \text{ K}$

$$\text{i. } N_0 = 30 + 10\log_{10}(kT_0) = 30 + 10\log_{10}(1.38 \times 10^{-23} \times 295)$$

$$N_0 \approx -173.90 \text{ dBm/Hz}$$

$$\text{ii. Thermal noise power} = N_0 + 10\log_{10}(B) = -173.90 + 10\log_{10}(10 \times 10^6) = -173.90 + 70.00$$

$$\text{Thermal noise power} \approx -103.90 \text{ dBm}$$

$$\text{iii. Noise floor} = \text{Thermal noise power} + NF = -103.90 + 7$$

$$\text{Noise floor} \approx -96.90 \text{ dBm}$$

$$\text{iv. Receiver sensitivity} = \text{Noise floor} + \text{required SNR} = -96.90 + 9$$

$$\text{Rx sensitivity} \approx -87.90 \text{ dBm}$$

(d) Maximum Allowable Path Loss (MAPL)

$$\begin{aligned}
 \text{MAPL} &= \text{EIRP} + G_{RX} - L_{RX} - \text{Rx sensitivity} - L_{\text{body}} - L_{\text{bldg}} - \text{Margin} \\
 &= 20.97 + 16 - 1 - (-87.90) - 2 - 6 - 9
 \end{aligned}$$

$$\text{MAPL} \approx 106.87 \text{ dB}$$

(e) Coverage Radius and Area (Log-Distance Model, $n = 3.0$, $d_0 = 100 \text{ m}$)

$$\begin{aligned}
 PL(d) &= PL(d_0) + 10n \cdot \log_{10}(d/d_0) \\
 106.87 &= 77.55 + 10(3.0) \cdot \log_{10}(d/100) = 77.55 + 30 \cdot \log_{10}(d/100) \\
 \log_{10}(d/100) &= (106.87 - 77.55)/30 = 0.9776 \\
 d &= 100 \times 10^{0.9776}
 \end{aligned}$$

$$\text{i. } R \approx 949.6 \text{ m } (\approx 0.950 \text{ km})$$

$$\text{Area} = (3\sqrt{3}/2)R^2 = 2.598 \times (0.9496)^2$$

$$\text{ii. Area} \approx 2.343 \text{ km}^2$$

(f) Effect of a 3 dB Improvement in Receiver Sensitivity

$$\begin{aligned}
 \text{MAPL}_{\text{new}} &= 106.87 + 3 = 109.87 \text{ dB} \\
 \log_{10}(d/100) &= (109.87 - 77.55)/30 = 1.0776 \Rightarrow d \approx 1195.5 \text{ m} \\
 \text{Area}_{\text{new}} &= 2.598 \times (1.1955)^2 \approx 3.713 \text{ km}^2
 \end{aligned}$$

$$R_{\text{new}} \approx 1.196 \text{ km; coverage area increases by } \approx 58.5\% \text{ (from } \approx 2.343 \text{ km}^2 \text{ to } \approx 3.713 \text{ km}^2).$$

Problem 3 — UMTS Handheld Link Budget (32 kbps, 15 users)

Body-loss convention: unlike Problem 2 above, this is a WCDMA/UMTS uplink problem, so it follows Module 3's canonical mobile-uplink EIRP formula (p.16): "EIRP = Mobile TX power + antenna gain – body loss" — the same convention Tutorial 1 Problems 3 & 4 use. Body loss is subtracted directly in the EIRP line, not added later. (In earlier problem sets this device was a laptop data card with 0 dB body loss, so the subtraction was invisible; here the device is handheld, so it isn't.)

Step 1: Mobile EIRP

$$\text{EIRP} = P_{\text{TX}} + G_{\text{TX}} - L_{\text{body}} = 23 + 1 - 2$$

$$\text{EIRP} = 22 \text{ dBm}$$

Step 2: Processing Gain

$$G_p = 10 \log_{10}(M_{\text{CPS}}/R_s) = 10 \log_{10}(3.84 \times 10^6 / 32 \times 10^3) = 10 \log_{10}(120)$$

$$G_p \approx 20.79 \text{ dB}$$

Step 3: Receiver Noise Power

$$\text{RNP} = N_0 + 10 \log_{10}(M_{\text{CPS}}) + \text{NF} = -174 + 10 \log_{10}(3.84 \times 10^6) + 5.5 = -174 + 65.84 + 5.5$$

$$\text{RNP} \approx -102.66 \text{ dBm}$$

Step 4: Total Noise Interference

$$\text{TNI} = \text{RNP} + \text{Interference margin} = -102.66 + 3.5$$

$$\text{TNI} \approx -99.16 \text{ dBm}$$

Step 5: Effective Sensitivity (with N = 15 users)

Since the problem states "considering 15 users", include the $10 \log_{10}(N)$ term:

$$S_e = 10 \log_{10}(N) + \text{TNI} - G_p + E_b/N_0$$

$$= 10 \log_{10}(15) + (-99.16) - 20.79 + 2.5 = 11.76 - 99.16 - 20.79 + 2.5$$

$$S_e \approx -105.69 \text{ dBm}$$

(b) Maximum Allowable Path Loss

$$\text{MAPL} = \text{EIRP} - S_e + G_{\text{BS}} - L_{\text{feeder}} - M_{\text{ff}} - M_{\text{sf}} - L_{\text{bldg}} + G_{\text{SHO}}$$

$$= 22 - (-105.69) + 16 - 2 - 4 - 7 - 13 + 2$$

$$\text{MAPL} \approx 119.69 \text{ dB}$$

(c) Coverage Radius

$$\text{MAPL} = 134.0 + 34.5 \log_{10}(R)$$

$$\log_{10}(R) = (119.69 - 134.0)/34.5 = -0.4148$$

$$R = 10^{-0.4148} \approx 0.385 \text{ km} (\approx 385 \text{ m})$$

Problem 4 — WCDMA Uplink: 256 kbps Video in Dense Urban

Part (a) Step-by-Step Link Budget

i. $\text{EIRP} = P_{\text{TX}} + G_{\text{TX}} - L_{\text{body}} = 22 + 1.5 - 3.5$

$$\text{EIRP} = 20.0 \text{ dBm}$$

ii. Receiver Noise Power

$$P_N = N_0 + 10 \log_{10}(M_{\text{CPS}}) + \text{NF} = -174 + 10 \log_{10}(3.84 \times 10^6) + 5.2 = -174 + 65.84 + 5.2$$

$$P_N \approx -102.96 \text{ dBm}$$

iii. Total Noise-Plus-Interference Power

$$I_{\text{total}} = P_N + \text{MI} = -102.96 + 8$$

$$I_{\text{total}} \approx -94.96 \text{ dBm}$$

iv. Processing Gain

$$G_p = 10\log_{10}(3.84 \times 10^6 / 256 \times 10^3) = 10\log_{10}(15)$$

$$G_p \approx 11.76 \text{ dB}$$

v. Effective Receiver Sensitivity (no user count given $\Rightarrow$ base formula, no $10\log_{10}(N)$ term)

$$S_e = I_{\text{total}} + E_b/N_0 - G_p = -94.96 + 5.2 - 11.76$$

$$S_e \approx -101.52 \text{ dBm}$$

vi. Maximum Allowable Path Loss

$$\text{MAPL} = \text{EIRP} - S_e - M_{\text{sf}} + G_{\text{SHO}} = 20.0 - (-101.52) - 9.5 + 1.8$$

$$\text{MAPL} \approx 113.82 \text{ dB}$$

vii. Cell Radius (given model $\text{PL}_{\text{max}} = 136.5 + 35.8 \log_{10}(R)$)

$$\log_{10}(R) = (113.82 - 136.5) / 35.8 = -0.6335$$

$$R \approx 0.232 \text{ km} (\approx 232 \text{ m})$$

Part (b) Quantitative Impact Analysis

$$\text{Baseline coverage area: } A = 2.598 \times (0.232)^2 \approx 0.140 \text{ km}^2$$

(i) Bit Rate Increased to 512 kbps (same E_b/N_0)

$$\text{New processing gain: } G_p' = 10\log_{10}(3.84 \times 10^6 / 512 \times 10^3) = 10\log_{10}(7.5) \approx 8.75 \text{ dB}$$

$$\text{New } S_e' = -94.96 + 5.2 - 8.75 \approx -98.51 \text{ dBm}$$

$$\text{New MAPL}' = 20.0 - (-98.51) - 9.5 + 1.8 \approx 110.81 \text{ dB}$$

$$\log_{10}(R') = (110.81 - 136.5) / 35.8 = -0.7178 \Rightarrow R' \approx 0.192 \text{ km}$$

$$A' = 2.598 \times (0.192)^2 \approx 0.0953 \text{ km}^2$$

$$R' \approx 192 \text{ m; coverage area shrinks by } \approx 32.1\% \text{ (from } \approx 0.140 \text{ km}^2 \text{ to } \approx 0.0953 \text{ km}^2).$$

(ii) Interference Margin Rises to 11 dB (higher network load, bit rate back to 256 kbps)

$$\text{New total noise + interference: } I''_{\text{total}} = -102.96 + 11 = -91.96 \text{ dBm}$$

$$\text{Processing gain unchanged: } 11.76 \text{ dB}$$

$$\text{New } S_e'' = -91.96 + 5.2 - 11.76 \approx -98.52 \text{ dBm}$$

$$\text{New MAPL}'' = 20.0 - (-98.52) - 9.5 + 1.8 \approx 110.82 \text{ dB}$$

$$\log_{10}(R'') = (110.82 - 136.5) / 35.8 = -0.7175 \Rightarrow R'' \approx 0.192 \text{ km}$$

$$A'' = 2.598 \times (0.192)^2 \approx 0.0955 \text{ km}^2$$

$$R'' \approx 192 \text{ m; coverage area shrinks by } \approx 32.0\% \text{ (from } \approx 0.140 \text{ km}^2 \text{ to } \approx 0.0955 \text{ km}^2).$$

Trade-off Discussion

As in the earlier problem sets, both changes remove almost exactly the same amount (≈ 3 dB) from the link budget and produce almost identical area shrinkage ($\approx 32\%$). This is "cell breathing": loading a CDMA cell with more traffic or higher bit rates shrinks its effective radius, because the added interference (or added bit-rate demand) eats directly into the same link margin that otherwise buys coverage distance.